Shortest path algorithm

Michel Bierlaire

Solution of the practice quiz

As some arcs in the network have negative cost, it is important to verify that the problem is bounded during the course of the algorithm. Consider a network with n arcs and m nodes where $c \not\geq 0$. Consider an arbitrary pair of different nodes, and P the shortest path between the two. Then,

$$C(P) \geq (m-1) \min_{(i,j)} c_{ij},$$

because a simple path cannot contain more than $m-1$ arcs. In this network, $m = 6$ and $\min_{(i,j)} c_{ij} = -9$. Therefore, no shortest path has a cost lower than -45.

Each iteration of the algorithm is summarized in Table 1, where

- the first column reports the iteration number,
- the second column reports the content of set $\mathcal{S}$,
- the third column reports the node that is selected to be treated in the next iteration,
- all the other columns contain the labels of all the nodes in the network.

The algorithm is interrupted at iteration 14, as the label of node 2 is -50. As it is below the lower bound, it means that the problem is non bounded. Indeed, the cycle $2 \to 5 \to 3 \to 2$ has length $5 - 7 - 9 = -11$. And the cycle $2 \to 4 \to 5 \to 3 \to 2$ has length $7 + 4 - 7 - 9 = -5$. There is no shortest path.

Here we have $\lambda_2 < 0$ and

$$\lambda_2 < (m-1) \min_{(i,j) \in \mathcal{A}} c_{ij} = 5 \cdot (-9) = -45.$$

Iteration	S	i	λ_1	λ_2	λ_3	λ_4	λ_5	λ_6
0	$\{1\}$	1	0	∞	∞	∞	∞	∞
1	$\{2,3\}$	2	0	-1	3	∞	∞	∞
2	$\{3,4,5\}$	3	0	-1	3	6	4	∞
3	$\{2,4,5\}$	2	0	-6	3	6	4	∞
4	$\{4,5\}$	5	0	-6	3	1	-1	∞
5	$\{3,4,6\}$	3	0	-6	-8	1	-1	-3
6	$\{2,4,6\}$	2	0	-17	-8	1	-1	-3
7	$\{4,5,6\}$	5	0	-17	-8	-10	-12	-3
8	$\{3,4,6\}$	3	0	-17	-19	-10	-12	-14
9	$\{2,4,6\}$	2	0	-28	-19	-10	-12	-14
10	$\{4,5,6\}$	5	0	-28	-19	-21	-23	-14
11	$\{3,4,6\}$	3	0	-28	-30	-21	-23	-25
11	$\{2,4,6\}$	2	0	-39	-30	-21	-23	-25
12	$\{4,5,6\}$	5	0	-39	-30	-32	-34	-25
13	$\{3,4,6\}$	3	0	-39	-41	-32	-34	-36
14	$\{2,4,6\}$	-	0	-50	-41	-32	-34	-36

Table 1: Iterations of the shortest path algorithm

Thus a negative cycle is detected and the algorithm fails as the problem is unbounded. Non negative cycle is indeed an assumption to allow the algorithm to finish after a finite number of iterations.